

Civics Group	A Level Index Number	Name (use BLOCK LETTERS)
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H2

**ST. ANDREW'S JUNIOR COLLEGE
2021 JC2 PRELIMINARY EXAMINATIONS**

H2 BIOLOGY**9744/03****Paper 3 Mark Scheme**

Thursday

16th September 2021Total 2 hours for
both Set A & B**READ THESE INSTRUCTIONS FIRST**

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Set A (Structured Question 1)

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Set B**(Structured Questions 2 & 3)**

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

(Essay Question 4 & 5)

Answer **one** essay question (parts a and b).

Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

For Examiners' Use	
Set A	
1	/35
Set B	
2	/5
3	/10
4 or 5	/25
Total	/75

Conceptual error (C)	Data Quoting (D)	Expression (E)	Misreading the question (Q)

This document consists of **xx** printed pages.

[Turn over]

Answer **all** questions.

QUESTION 1

Many diseases are linked to components of the immune system.

T-cell lymphoma is a rare form of cancerous lymphoma affecting T lymphocytes. Lymphoma arises mainly from the uncontrolled proliferation of T lymphocytes and can become cancerous.

(a) Cancer is described as a multi-step process. Explain how mutations in two main groups of critical genes may lead to uncontrolled proliferation in T lymphocytes.

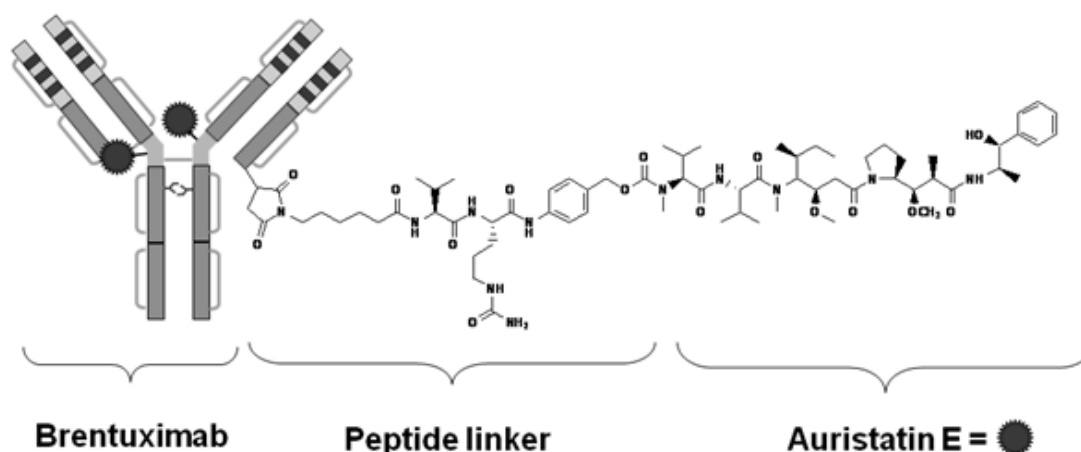
.....[4]

1. loss of function mutation of tumour suppressor gene
/ mutations must affect both/two alleles in a cell's genome;
to produce **non-functional** tumour suppressor protein;
2. loss of **arrest** of cell division, loss of ability for DNA repair and loss of **apoptosis**
3. gain of function mutation of proto-oncogene
/ only need one mutated allele;
to produce **abnormal/hyperactive** proto-oncogene protein;
4. lead to stimulation of the cell cycle / cell keeps dividing;
5. allowing for subsequent **accumulation** of many other mutations (resulting in uncontrolled proliferation of T lymphocytes);

- (b) Possible treatment for lymphoma might include antibodies attached to chemotherapy drugs (an antibody-drug conjugate). For example, Brentuximab vedotin is an antibody attached to a chemotherapy drug.

This antibody specifically recognises CD30 found on lymphoma cells and acts like a “homing” signal, bringing the chemotherapy drug to lymphoma cells, where it enters the cells and kills them.

Fig. 1.1 shows the structure of this antibody-drug conjugate and how the drug is attached to the antibody.



Modified from http://www.pharmatimes.com/Article/13-05-15/FDA_to_review_extended_use_of_Seattle_Genetics_Adcetris.aspx

Fig .1.1

Auristatin E is an anti-mitotic agent which inhibits cell division by blocking the polymerisation of tubulin, a protein monomer of spindle fibres.

- (i) With reference to Fig 1.1, identify the region on the antibody that helps the antibody to achieve its “homing” function. Explain how this “homing” is achieved.

.....[2]

1. [Region] variable region / antigen binding site;
2. [Explain] Ref. variable region binds to **{CD30/receptor/glycoprotein}** found on lymphocytes due to **complementary 3D conformation**;

- (ii) Suggest how Auristatin E can help to curb T cell lymphoma.

.....[2]

Ref. any function of spindle fibres in mitosis in negative phrasing

1. e.g. **stops** the **alignment** of chromosomes at the metaphase plate / equator; during metaphase of mitosis;
/ e.g. **prevents** the **separation** of sister chromatids; during anaphase;
2. T lymphocyte **cannot undergo successful** mitosis and subsequent **cell division**; (thus, stopping uncontrolled proliferation);
3. AVP on mechanism on how Auristatin E might affect tubulin assembly

An investigation was conducted to determine the effectiveness of Brentuximab vedotin (antibody and auristatin E conjugate) treatment in humans.

A lymphoma cell line was incubated *in vitro* with Brentuximab vedotin (antibody and auristatin E conjugate).

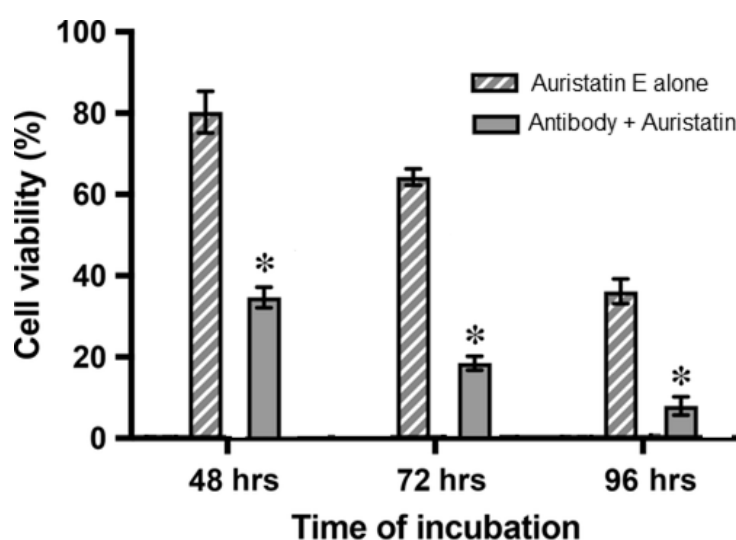
Another similar lymphoma cell line was incubated *in vitro* with auristatin E alone (unbound to antibody).

The investigation was done twice, with 9 replicates for each duration of incubation.

Cell viability (% of cells that survive) was assessed and expressed relative to untreated control in Fig. 1.2. The data for the control setup (for untreated cells) was not shown in the figure.

Data are presented as mean $\pm$ S.D. (standard deviation).

Statistical differences between the two groups of cells are marked by asterisk ($P < 0.01$).



Tip: Understand the set up. When more cancer cells die, the more effective the treatment!

Modified from Schonberger et. al, 2017, Journal of Cellular and Molecular Medicine

Fig. 1.2

(iii) Based on the information given, comment on the effectiveness of the Brentuximab vedotin (antibody and auristatin E) treatment in humans and discuss on the confidence in the presented data.

- [4]
1. The treatment is effective as shown by the consistently **lower** % of cell viability in antibody+auristatin E treatment compared to auristatin E alone;
 2. + quote of **any one time duration**
e.g. at 48h, % viability for antibody + auristatin E is lower at 35% (accept 34-38%) vs 80% in auristatin E lone;
/ at 72h, % viability for antibody + auristatin E is lower at 19% (accept 18%) vs 62% (accept 63 % and 64%) in auristatin E lone;

/ at 96h, % viability for antibody + auristatin E is lower at 5% (accept 4 - 6%) vs 38% (accept 36-39%) in auristatin E lone;

There is confidence in the **data** due to:

3. **High {reproducibility / reliability}** from performing of {**1 more** repeat experiment / **9 replicates**} done per experiment;
4. [Increased confidence] Ref. comparison to **untreated controls** in cell viability assays;
5. Ref. **statistical test analyses** revealed a **significant difference** between both treatments;
6. Ref. relatively **small standard deviation** for all groups, suggesting of a **consistency** in results **with respect to the mean**;

There is lack of confidence in the **data** of treatment due to:

4. Ref. data for untreated controls not shown, to allow better comparison

There is lack of confidence in the **efficacy** of treatment due to:

7. Experiment **performed on cell lines *in vitro***, need more clinical studies on humans;
8. AVP

(c) Lymphocytes in the human body, have a limited lifespan. These cells are continually replaced by the division of **lymphoid** stem cells. Fig. 1.3 shows that when a stem cell divides, two daughter cells are produced. One remains as a stem cell, whilst the other divides and differentiates to form specialised cells.

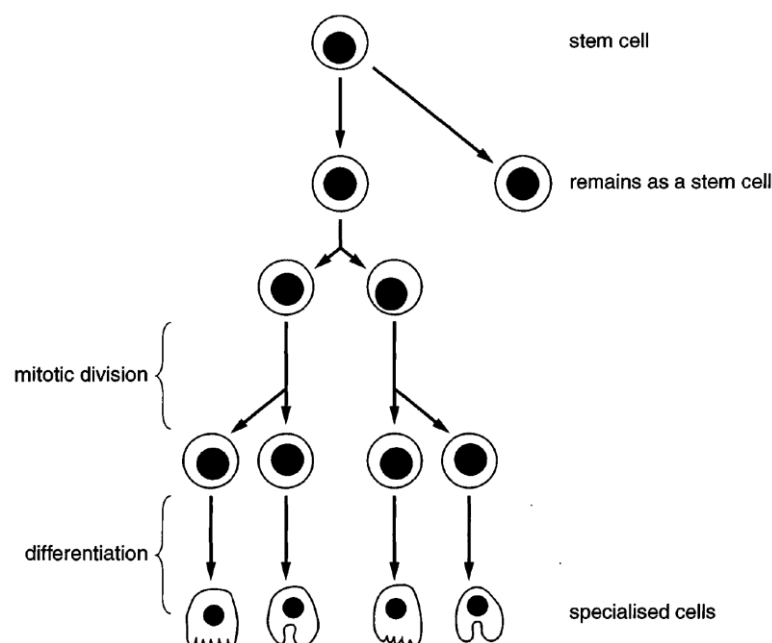


Fig. 1.3

(i) The specialised cells are genetically identical to the stem cells from which they originated. Suggest how a cell differentiates to form a specialised cell.

..... [2]

1. Ref. **differential gene expression** / expression and silencing of different genes;
2. Leading to **different gene products** that **contributes to the function and structure** of the cell; [Reject: leads to differentiation to form a specialised cell as this is already in pre-amble]
3. AVP: Ref. **cell signalling** due to external signals

(ii) Other than the location on where to find these stem cells, describe two differences between lymphoid stem cells and embryonic stem cells.

..... [2]

Feature	Lymphoid stem cells	Embryonic stem cells
Potential	<u>Multipotent</u> ; able to differentiate into limited cell types, of specific lineage;	<u>Pluripotent</u> ; able to differentiate into almost all cell types except extra-embryonic structures;
Example of differentiated cells / Function	Lymphoid stem cells differentiate into 3 types of white blood cells – <u>B and T lymphocytes and natural killer cells</u> / For the daily replacement of worn out blood cells from {normal wear and tear / fighting infections in diseases / from injury} ;	Give rise to various organs in organism (during development);

(iii) Embryonic stem cells are extracted from human embryos 4-5 days after fertilisation, when the embryo consists of 50 to 150 cells.

Suggest one ethical concern about the use of embryonic stem cells.

..... [1]

[Specific ethical issues associated with the use of ES cells]

1. **Destruction** of embryos donated by individuals undergoing IVF as they are **potential human lives** ;
2. Intentionally creating and **destroying embryos** during SCNT **violates respect** for human **lives** ;
3. **Medical risks** of **oocyte retrieval** from the ovaries for SCNT as it is an invasive procedure ;

4. **Reproductive interest** of women undergoing fertility treatment may be compromised if they consent to donate any of their **oocytes** for research purposes
[Any 1]

(d) Severe combined immunodeficiency (SCID) is a disease that affect the development and maturation of T and B lymphocytes.

There are two general types of SCID, both of which are caused by recessive alleles.

Fig. 1.4 shows a family pedigree including some people with a type of SCID.

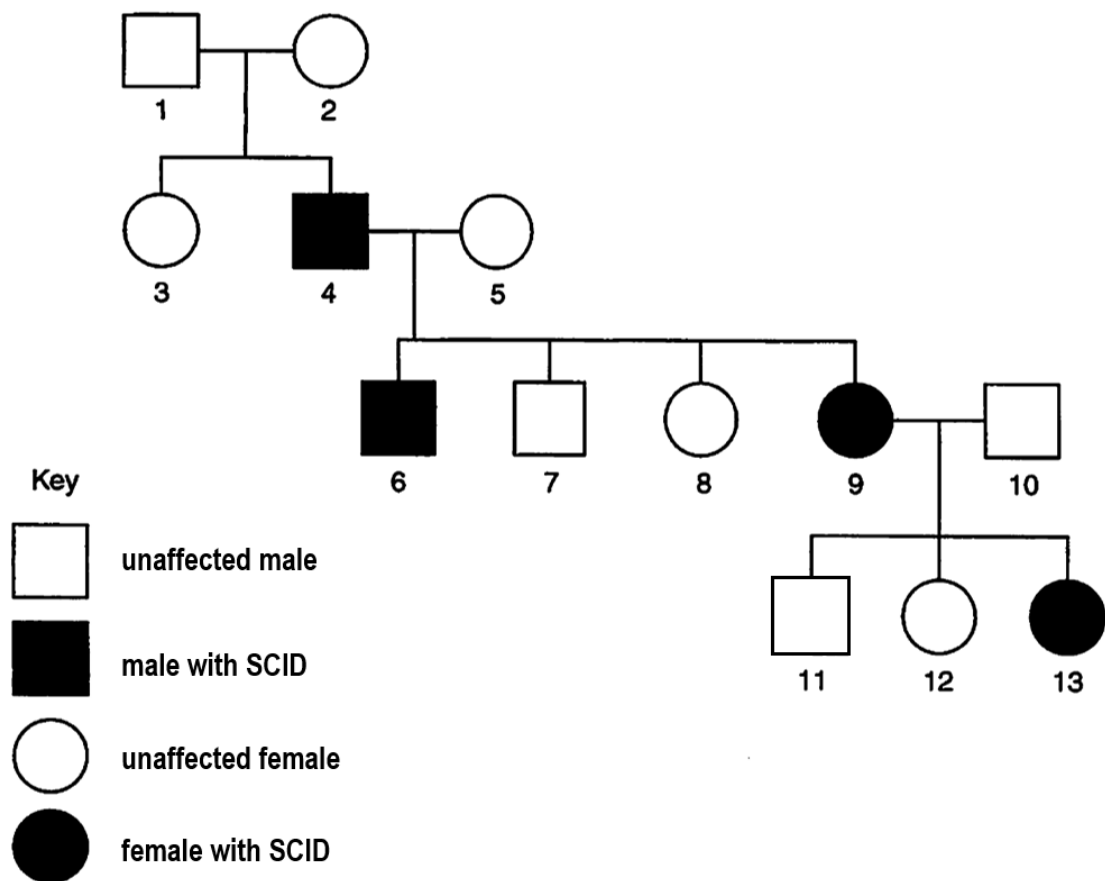


Fig. 1.4

(i) Explain what is meant by the term allele.

.....[2]

1. Allele: **alternative form** of a gene that occupy the same gene locus on homologous chromosomes; [Reject: "altered" (I.E means mutated)]
2. (after gene expression) causes slightly **different phenotypes** for the same characteristic due to a slight **difference in DNA sequence**;

(ii) Explain, with evidence in Fig. 1.4, whether the disease is X-linked.

.....[3]

1. [State] Not X-linked / Is autosomal; [Compulsory point]
[Pattern for X-linked recessive] If X linked recessive, all **diseased mothers** (X^aX^a) will **have** (only) **diseased sons** (X^aY) ;
2. [Explain] Ref. son of diseased mother will **inherit X chromosome** (X^a) containing disease allele (only) from mother;
3. [Quote evidence for not X-linked] Not observed in pedigree tree as, Individual 9 is **diseased** but her son individual 11 is **not diseased**; [Compulsory point!]
4. OVP

OR

1. [State] Not X-linked / Is autosomal; [Compulsory point]
2. [Pattern[#] for X-linked recessive] If X linked recessive, all **diseased daughters** (X^aX^a) will **have** (only) **diseased father** (X^aY) ;
3. [Explain] Ref. daughter of diseased father will **inherit an X chromosome** (X^a) containing disease allele (only) from father (and father only has one X chromosome) ;
4. [Quote evidence for not X-linked] Not observed in pedigree tree as, Individual 13 is **diseased** but her father individual 10 is **not diseased**; [Compulsory point]
5. OVP

OR

1. [State] Not X-linked / Is autosomal;
2. [Explain] If autosomal, we should observe approximately **equal** number of **males**, and **females affected** ;
3. [Explain] Ref. equal chance of **inheriting dominant or recessive allele from** (either) **parents** due to random fusion of gametes ;
4. [Quote evidence for autosomal]; Observed in pedigree tree as, ref. affected male individuals 4, 6 vs affected female individuals 9,13 ; [Compulsory point]
5. OVP

A southern blot analysis was performed to investigate the type of alleles of this gene which causes SCID in a family branch consisting of individuals 1, 2, 3 and 4.

Fig. 1.5 shows the steps involved in this analysis.

Steps of Southern Blot analysis

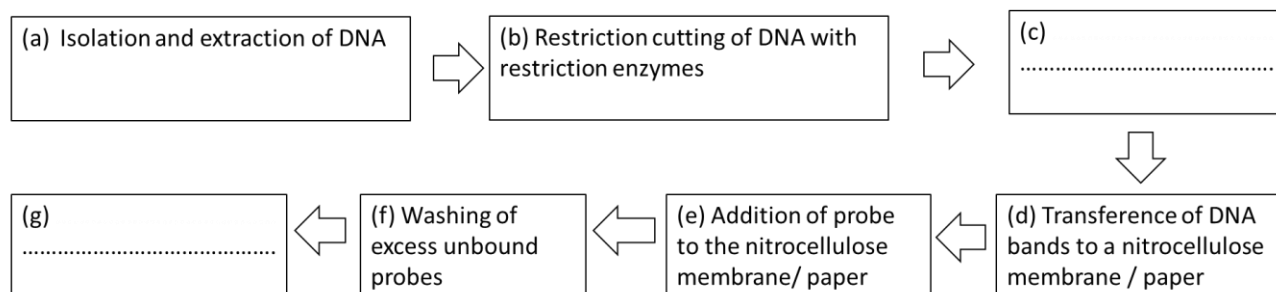


Fig. 1.5

(iii) Fill in the blanks for steps (c) and (g) as shown in Fig. 1.5.

.....[1]

BOTH for 1M:

Step (c): Gel electrophoresis

Step (g): X ray autoradiography (Allow for description)

Fig. 1.6 shows the restriction map of the dominant allele and recessive allele. Arrows represent restriction sites which restriction enzyme *Bam*HI recognises and binds. The size of the DNA fragments after restriction cutting is indicated in kb.

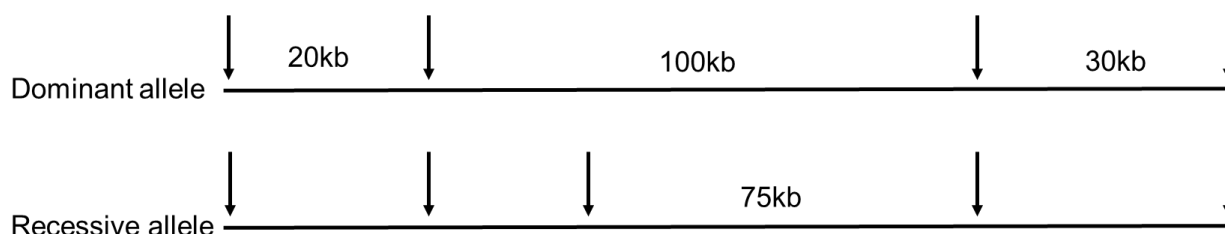


Fig. 1.6

A zoomed in view on each restriction site shows the following in Fig. 1.7.



Fig. 1.7

(iv) With reference to Fig. 1.6 and Fig. 1.7, describe and explain the mode of action of restriction enzyme *BamHI*.

-[3]
1. [Compulsory point] [Active site + Quote Fig 1.7] *BamHI* has an active site with **3D conformation complementary** to the restriction site of the DNA ; comprising of 5' GGATCC 3' and 3' CCTAGG 5' [**ACCEPT**: quote of **any** 1 strand] on both strands;
 2. [Mode of enzyme specificity]
(‘Lock and key’ hypothesis) - conformation of substrate is **fully complementary** to the 3D conformation of the active site of enzyme ;
OR
(Induced fit hypothesis) - initial binding of substrate to enzyme causes a **change in the 3D conformation** of the enzyme which leads to **more effective binding** at the active site of the enzyme ;
 3. [Activation energy] Ref. *BamHI* enzyme **lowers** the activation energy ;
by inducing **stress** in **bonds** of substrate / **Increases** substrate **reactivity** ;
(Reject: Serves as a template to position (different) substrate molecules in the correct orientation for catalysis; due to nature of the Q’s context)
 4. [Contextualise enzyme’s action] Ref. catalytic residues of enzyme active site will catalyse the **cleavage** of phosphodiester bond/s at the restriction site;
 5. [Contextualise products - Quote Fig 1.6] Ref. cleaving dominant allele produces product of length 20kb, 30kb and 100kb / cleaving recessive allele produces product of length 20kb, 25kb, 30kb and 75kb / after action of restriction enzyme, this forms fragment length of 100kb for dominant allele and fragments length of 25kb and 75kb for recessive allele) ;

(v) Compare between probes (an example is the one used in step (e) in Fig. 1.5) and primers used in Polymerase Chain Reaction (PCR).

.....[2]
2M = Max 1 Similarity + Max 1 Difference awarded

[Similarity] **Any 1**

1. Both are **complementary** in **sequence** to **target** gene;
2. Both are single stranded polynucleotide chains;
3. Both are made of DNA;
4. AVP e.g. both are **artificially** synthesized;

[Differences] **Any 1**

5. [Radioactivity] Probe is **radioactive** but not primers;
6. [Function] Probe is used to **detect** the **presence** of target genes but primers are used to **start off replication** of the target gene;
7. AVP

(vi) Suggest the significance of step (f) in Fig. 1.5.

.....[1]

Step (f) involves washing off of unbound probes,

1. [Significance] This will allow the **clear visualisation / detection** of DNA bands which are actually **complementary / bound** by the probe;

ACCEPT: **lower** chance of getting **false positive** results (from unbound probes) ;

(Further elaboration for learning:

Probes are radioactive, so any *unbound* probes which remain on the nitrocellulose paper due to lack of washing will be detected and will also darken the X ray film. Thus, this may result in false positive results.)

Fig. 1.8 shows the results of step (g) in Fig. 1.5.

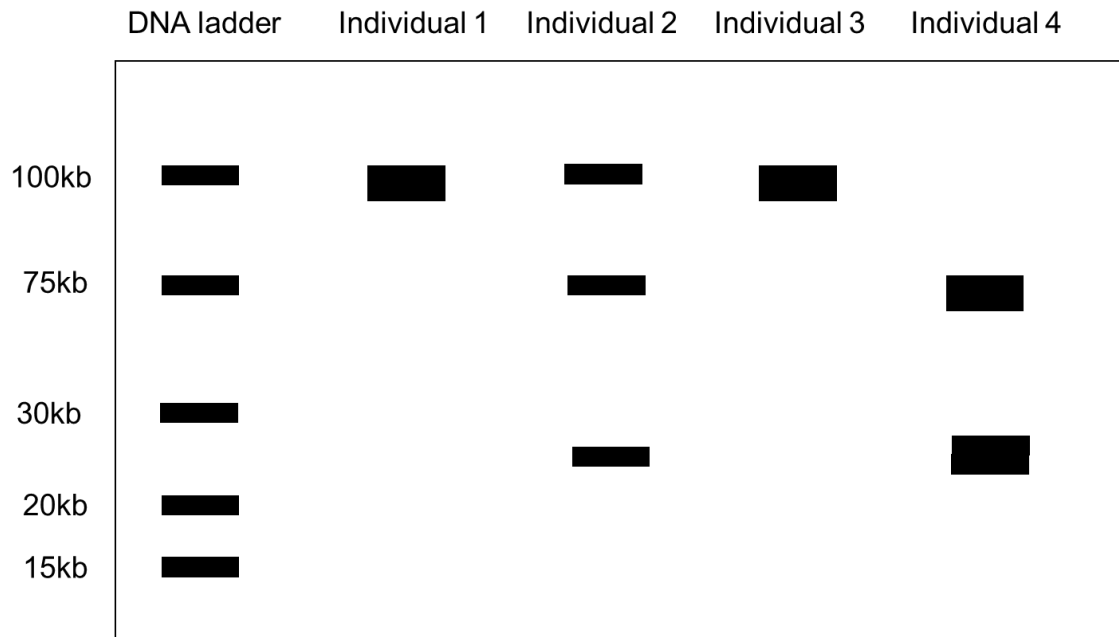
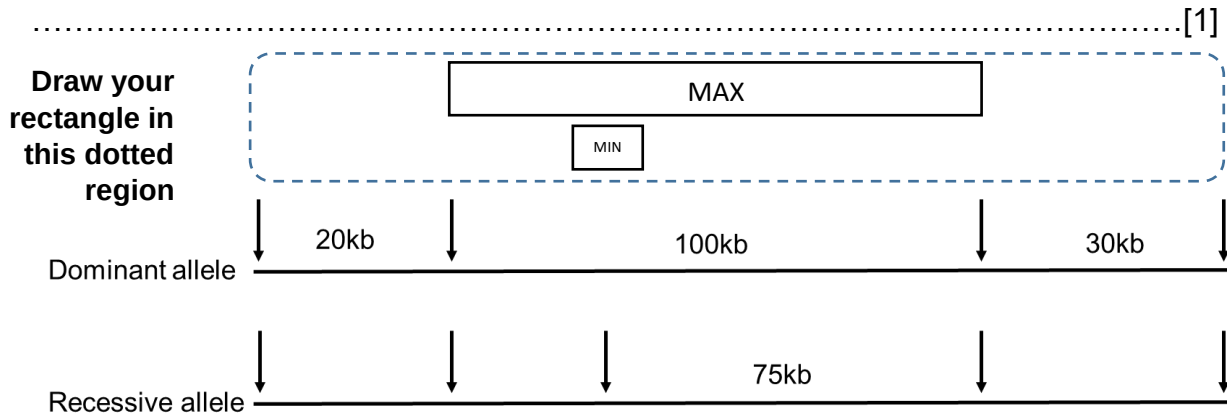


Fig. 1.8

(vii) With reference to Fig. 1.8, and using a rectangular box () to represent the probe, suggest a suitable region which the probe is binds to.

Draw your rectangular box in the **dotted region** of the diagram below.



1. Any region that overlaps the 25kb and 75kb fragments, BUT not exceeding restriction site flanking 100kb;

(viii) With reference to Fig. 1.4 and the results in Fig. 1.8, explain in detail, how this condition might have arisen in individual 4.

.....[3]

1. [Background context deduced from Fig 1.4 & Fig 1.8]
Individual 4 can only **inherit 1 recessive allele** from individual 2;
2. [Reason for cause of condition in Individual 2]
The **other** recessive allele could have arisen from **{mutation in Individual 1's gametes / mutation in (1 of the alleles of blood stem cells of) individual 4 after fertilisation}** ; [REJECT: double nondisjunction scenario]

[Effects of DNA mutation to cause the condition]
3. Different **amino acid sequence** coded for ;
different R groups of the amino acids forms **different interactions** ;
[Accept alternative effects: frameshift effect / premature stop codon]
4. Change in **3D conformation** of the **protein**;
loss of function of the protein, thus causing leading to lack of mature / developed T and B lymphocytes ;
5. OVP [Deduction of Genotype of Parents]
From Fig. 1.8, Individual 1's genotype is **homozygous dominant /AA** (as a thick 100kb band is observed) while Individual 2's genotype is **heterozygous / Aa** (as thin 25kb, 300kb and 100kb bands were observed) ;
6. OVP [Alternative reason as opposed to MP2]
E.g. Individual 4's biological father was **another man** and not individual 1 ;

(ix) Individual 12 [Aa] in Fig. 1.4 settled down with a male with no family history of SCID. [AA]

Calculate the probability of this couple having a son who is a carrier of the disease. Show your working clearly.

.....[2]

1. Probability (both son and a carrier) = $\frac{1}{2} \times \frac{1}{2}$

2. $= \frac{1}{4}$ [Accept: 0.25]

[Total 35 marks]

QUESTION 2

(a) Researchers of the International AIDS Vaccine Initiative (IAVI) and Scripps Research have tested a new vaccine approach designed to prevent HIV infections.

The vaccine consists of the engineered HIV Envelope (Env) proteins linked to a spherical protein structure. The engineered proteins are designed to engage with specific B lymphocytes of the immune system to produce a certain type of antibody that is capable of neutralizing a wide range of HIV variants.

(i) Classify the type of immunity conferred by this mode of vaccination in table 2.1 below.

.....[1]

Table 2.1

Type of immunity	✓ if applicable
Passive	
Active	✓
Natural	
Artificial	✓

(b) Naïve B lymphocytes express **membrane-bound** immunoglobulin M (IgM) in the form of B cell receptors, while plasma cells express the **secreted** form of IgM. The secreted IgM lack the transmembrane domain which embeds the IgM / B cell receptor in the cell surface membrane.

These structural differences is due to the process of alternative splicing of the same pre-mRNA coded by the same gene.

Both the B cell receptor and the IgM secreted by the same cell line should recognise the same antigen shape.

Describe how the plasma membrane of naïve B cells hold onto the B cell receptors.

.....[2]

1. hydrophobic interactions which are formed between the **{hydrophobic / non-polar} hydrocarbon chains of fatty acids of the phospholipids** and the **{hydrophobic / non-polar} R groups of the amino acids** on surface of proteins (in contact with the phospholipids molecules) ;
2. ionic and hydrogen bonds formed between the **{charged phosphate group / polar phosphate heads} of the phospholipids** and the **{charged / polar} R groups of the amino acids** on the surface of the protein;

(c) Indicate true or false for each of the following statements.

.....[2]

[Any 2 correct for 1m]

Accept: T/F

Reject: tick ✓ and cross X

No.	Statement	True/False
1	Somatic hypermutation occurs to allow higher affinity to the same antigen	True
2	Gene segments (e.g. v, d, j segments) coding for variable regions of antibodies are only found in the genome of B lymphocytes.	False
3	<i>Mycobacterium tuberculosis</i> is a bacterium pathogen which infects the macrophages in the alveoli after transmission through air-borne particles.	True
4	Primary infection only involves the innate immune system; secondary infection involves the specific immune system.	False

[Total 5 marks]

QUESTION 3

Sehuencas water frogs (*Telmatobius yuracare*) are endemic to Bolivia. It was once believed that this frog was extinct in the wild due to predation.

Recently, two distinct groups of Sehuencas water frogs were rediscovered in a big Bolivian forest. One group of water frogs is predominantly green in skin colour, while the other group is predominantly brown.

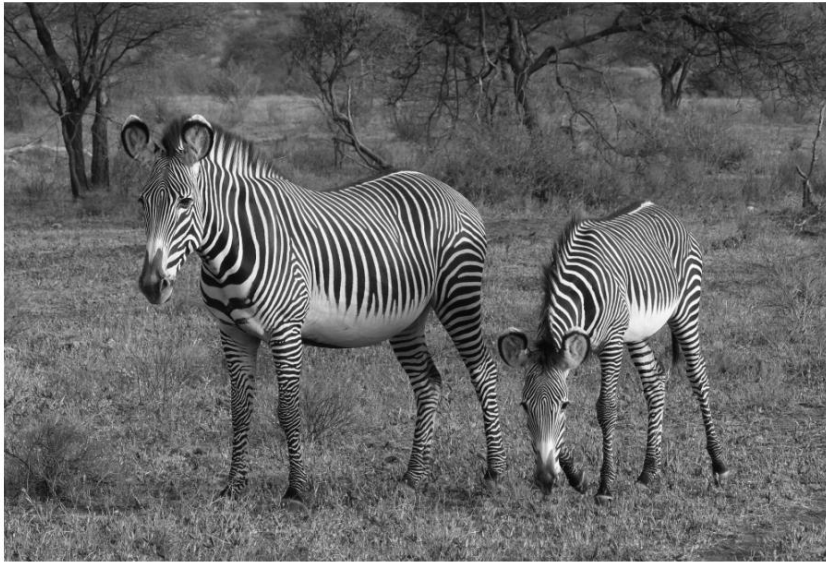
Despite being found in the same forest, each group is observed to be mating at different times of the year.

(a) Explain how the skin colour differences between the two groups of water frogs may have arisen.

.....[5]

1. **Temporal isolation** occurring between two groups of water frogs
[Reject:
 - sympatric speciation due to two groups are still the species
 - sexual selection
 - behavioural isolation]
 2. There is **disruption of gene flow** / no interbreeding between water frog populations ;
 3. **Genetic variations** exist among the water frog populations (due to spontaneous mutations);
 4. **Different selection pressures** in different habitats;
 5. **Ref. 1 e.g.** of selection pressure: e.g. shaded regions offering protection for brown frogs vs in the open, having more green plants, better camouflage for green frogs; [Accept any feasible example] ;
 6. more Individuals with a **selective advantage** in the particular environment **survived** till reproductive age; and pass on their alleles to offspring ; [**REJECT**: characteristics/traits passed on to offspring]
 7. **Change in allele frequency** of gene pool **over time** ;
 8. Other evolutionary agents such as **genetic drift** / founder's effect / bottleneck effect) / **mutation** occur;
- [Max 5]

(b) The photograph shows two zebras.



(Source: Caroline Wilcox)

In the wild, the stripes of zebras are thought to reduce attack by biting flies.

Biting flies cause stress and disease to cattle (e.g. cows), reducing food production and causing financial losses for farmers.

An investigation was carried out to see if the presence of stripes on cattle reduced the number of biting flies on their bodies.

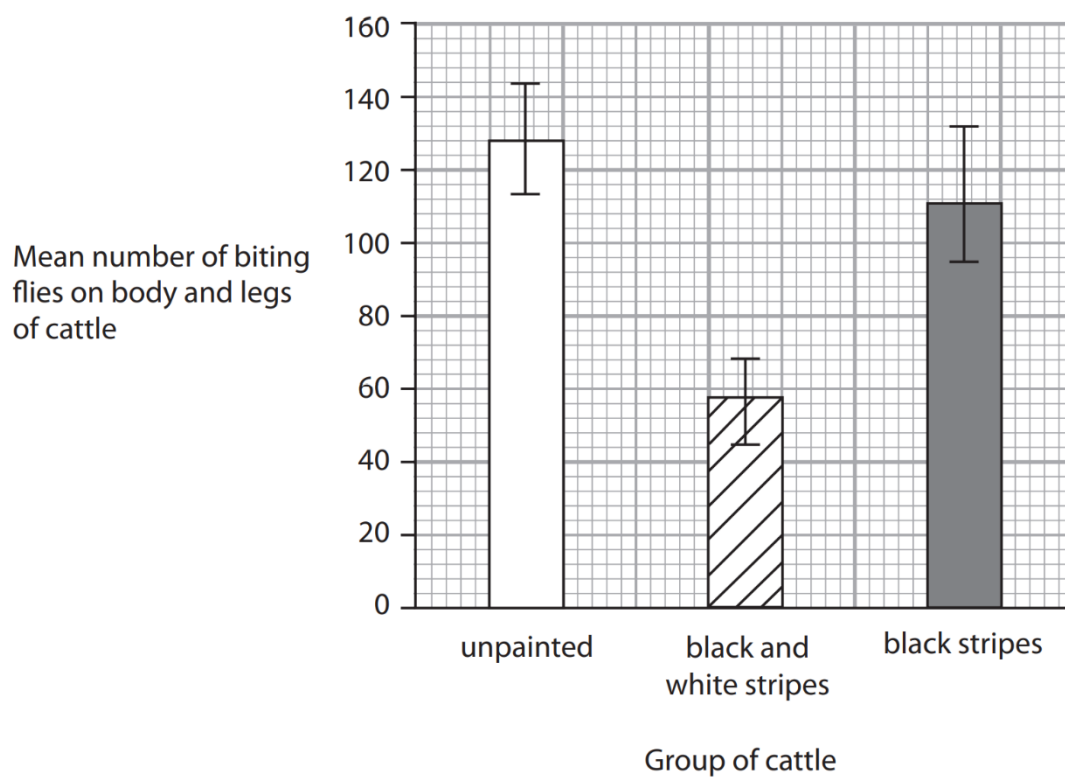
Three groups of black cattle were compared:

- unpainted cattle
- cattle painted with black and white stripes
- cattle painted with black stripes only

Cattle repel flies by flicking their tails, stamping their feet and skin twitching.

The frequencies of these fly-repelling behaviours were recorded.

The graph (Fig 3.1) and Table 3.1 show some of the results of this investigation.

**Fig. 3.1****Table 3.1**

Group of cattle	Mean frequencies of fly-repelling behaviours / number per hour		
	Flicking their tails	Stamping their feet	Skin twitching
unpainted	54	16	5
black and white stripes	42	10	8
black stripes	55	16	5

(i) Determine the effectiveness of painting stripes on cattle in reducing the number of biting flies on them.

Use the information in the graph (Fig. 3.1) and Table 3.1 to support your answer.

.....[4]
[state effects of painting stripes based on graph & table]

- 1 Painting stripes on cattle is **effective** in reducing the number of biting flies on them ;
- 2 but does not reduce significantly their flicking their tails, stamping their feet and skin twitching to repel flies ;

With regards to *[graph / Fig. 3.1]: any 2 [data quoting]*

- 3 *[graph/Fig. 3.1]* black and white stripes **reduce** the number of biting flies on body and legs more than **unpainted** ; AND
[D] 128 down to {57/58} / by {71/70} / by {55/56}% ;
- 4 black and white stripes **reduce** the number of biting flies on body and legs **more** than just **black stripes** ; AND
[D] 111 (Accept 112) down to {57/58} / by {53/54} / by {48/49}% ;
- 5 black stripes alone **reduce** the number of biting flies on body and legs **more** than **unpainted** ; AND
[D] 128 down to 111 (Accept 112) / by {16/17} / by {12.5/13.3}% ;

With regards to *[table 3.1]: any 2 [data quoting]*

- 6 black and white stripes have small effect on {flicking tail / stamping feet} compared to **unpainted**; AND
[D] {54 down to 42 / by 12 / by 22% (tail)} / {16 down to 10 / by 6 / by 38% (feet)} ;
- 7 black and white stripes have small increase on skin twitching ; AND
[D] 5 up to 8 / by 3 / by 60% compared to **unpainted**;
- 8 Ref. black stripes alone have no difference in { stamping feet / skin twitching} compared to **unpainted**;

(ii) Suggest a rationale for painting black stripes on the black cattle in the investigation.

-[1]
1. To ensure that any repelling of flies was not due to the smell of paint;
 2. AVP

[Total 10 marks]

Essay

Answer **one** question only in this section.

Write your answers on the lined paper provided at the end of this question paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 4 (a)** Describe how the level of mRNA of a particular gene across different cell types in an organism is controlled. Suggest the advantage(s) of each level of control. [12]

- (b)** Discuss how complementary base pairing, both within and between biomolecules, enables prokaryotic cells to carry out processes essential to life [13]

[Total: 25]

- 5 (a)** Many different polymers are formed in cells. A polymer is a chain of monomers. [12]

Describe the polymerisation of different types of biomolecules in a plant.

- (b)** Discuss how the complementary shapes between biomolecules enable eukaryotic cells to carry out their diverse processes [13]

[Total: 25]

SAJC Prelim 2021 – H2 Biology Paper 3 Essay Answers

Essay 4

- (a) Describe how the level of mRNA of a particular gene across different cell types in an organism is controlled. Suggest the advantage(s) of each level of control.

[12]

[A] DNA (Chromosomal) Level

Gene expression is switched on by:

1. **DNA demethylation** by DNA demethylase
2. **Histone acetylation** by histone acetyltransferase
3. Ref. **neutralize positive charge** of lysine residues on histone tails; unable to interact with linker DNA,
4. to **loosen the chromatin** structure; **allowing** for **transcription** (general transcription factors and RNA polymerase to access promoter)

Gene expression is switched off by:

5. **DNA methylation** by DNA methyltransferase ;
6. Ref. leads to **alter the shape** of the **promoter** sequence / **physical blocking** of general transcription factors / RNA polymerase;
7. Ref. **DNA methylation** recruits histone deacetylase
8. **Histone deacetylation** by histone deacetylase ;
9. allows lysine residues on histone tails to **interact** with linker DNA, [penalize once for lack of “lysine” or “linker”]
10. leading to **compacting** of chromatin structure which **prevents transcription**;

Advantage

11. **Idea of longer term switching genes** on and off to restrict active genes to those required (by the cell line), so **more efficient / less wasteful of resources**.

[B] Transcriptional Control

Gene expression is switched on by:

12. **Binding** of general transcription factors and RNA polymerase to the promoter to form transcriptional initiation complex for transcription initiation.
13. Activator binds to enhancer, thereby **increases rate** of transcription.
14. Ref. interaction with co-activator which then bind to the transcription initiation complex at the promoter, leading to **more efficient positioning of RNA polymerase / TIC**;

Gene expression is switched off by:

15. Repressor binds to silencer, and hence **prevent transcription** of the gene.
16. Ref. by **blocking** the binding of **activator** proteins to the control elements or to components of the transcription machinery

17. Ref. repressor binds to its **control element (silencer) which is within an enhancer**

Advantage

18. Ref. Rate of transcription / expression can be regulated (at this level), **to meet short term requirement of the cell.**

[C] Post-Transcriptional Control

19. Addition of (7-methylguanosine) cap to the **5' end** of the pre-mRNA ; **Cap-binding protein** binds to the 5' cap,
20. Addition of 3' Poly(A) tail added to the **3' end** of the pre-mRNA by poly(A) polymerase ; Poly(A) binding protein binds to the poly(A) tail;
21. (Proteins bound to 5' cap and 3' poly(A) tail) protect the mRNA from degradation by **exonucleases**, hence **increases the half-life of mRNA**;
22. Ref. half-life of mRNA is determined by the **length** of the poly(A) tail;

[Reject due to no relation to level of mRNA]

- *facilitate the export of mature mRNA from nucleus to cytosol.*
- *act as site of attachment for translational initiation factors to promote the binding of ribosomes to promote translation.*

[Reject] RNA splicing and Alternative splicing: affects type of mRNA formed but not affecting level/amount of mRNA.

Advantages

23. Ensures the stability of mRNA and hence allows the **regulation of levels of mRNA** present as a template for translation;

OWC [1]

At least 2 levels of control; with 2 correct mark points each (inclusive of advantage);

Reject: all points on prokaryotes because question was on different cell types, signifying of eukaryotes.

Examiner's comments

- Many candidates did not manage to adequately suggest the advantage of each level of control.
- Several answers also lack specific details (eg enzymes involved) when describing the process for each level of control.

4(b) Discuss how complementary base pairing, both within and between biomolecules, enables prokaryotic cells to carry out processes essential to life. [13]

A) Introduction

1. Complementary base pairing occurs in **nucleic acid-nucleic acid interactions** (held together by hydrogen bonds) ;
2. (H-bonds are) electrostatic forces of attraction between a hydrogen atom and a lone pair of electrons on an **electronegative** Nitrogen or Oxygen atom ;
3. Adenine (A) binds to Thymine (T) / Uracil (U) via 2 hydrogen bonds ; Cytosine (C) binds to Guanine (G) via 3 hydrogen bonds ;

B) Within a Nucleic Acid / I.E Molecular Structure

4. [General role] **Ref. to promoting stability** of the molecular structure ;
 - [**E.g.** In DNA – chromosome or plasmid]
5. Complementary base pairing / hydrogen bonds **between 2 DNA strands** in **chromosome / plasmid**;
6. **Ref. to role of storing genetic information** ;
 - [**E.g.** In tRNA]
7. Intra-molecular hydrogen bonding / complementary base pairing in tRNA allows tRNA to fold into a clover-leaf structure ;
8. **Ref. to role of tRNA – carries specific amino acids to the ribosome** for synthesis of polypeptide ;
 - [**E.g.** In rRNA]
9. Intra-molecular hydrogen bonding / complementary base pairing in rRNA allows rRNA to fold into a **precise 3D structure** (to complex with ribosomal proteins to form ribosome);
10. **Ref. to role of rRNA in ribosome - ribozyme/ peptidyltransferase** that catalyzes **peptide bond formation**;

C) Between 2 Nucleic Acid structures / I.E Process

11. [General role] **Ref.** to accurate **reading** of **template** in a process ;

- [**E.g.** During DNA replication (in Binary Fission)]

12. Important in DNA replication ; where complementary base pairing **between parental DNA strand** and **free deoxyribonucleotides** to synthesise the **daughter strand**;

13. **Ref.** to role of CBP for process above - to ensure **accurate transmission** of genetic information ;

14. Also important for the complementary base pairing **between parental DNA strand** and **free ribonucleotides** to synthesise the RNA primers;

15. **Ref.** to role of CBP for process above – RNA primer **provide existing 3'OH group** for DNA Polymerase III/I to add **deoxyribonucleotides to form daughter strand** / DNA Polymerase III/I cannot start from scratch ;

- [**E.g.** During DNA Proof-reading / repair]

16. Important in **DNA repair / proofreading**; where complementary base pairing **between original / intact DNA strand** and **free deoxyribonucleotides** to restore the **damaged / mutated strand** ;

17. **Ref.** to role of CBP for process above- to **preserve fidelity** of genetic information ; (note: importance of enzyme in achieving this role)

- [**E.g.** During Transcription]

18. Important in transcription ; where complementary base pairing **between template DNA / gene** and **free ribonucleotides** to synthesise the **(m)RNA product** ;

19. **Ref.** to role of CBP for process above - to ensure **correct sequence of codons / bases** in the RNA molecule for gene expression ;

- [**E.g.** During Translation]

20. Important in translation ; where complementary base pairing **between codons** on mRNA with anticodon on tRNA to synthesise the **polypeptide** ;

21. **Ref.** to role of CBP for process above - to ensure **correct sequence of amino acids** for functional protein ;

- [**E.g.** During Conjugation]

- 22.** Important in conjugation ; where complementary base pairing **between** single stranded F-plasmid from **donor/F+ cell** with **free deoxyribonucleotides** to synthesise the **a double stranded F-plasmid** in the **recipient / F- cell** (lagging strand synthesis);
OR
complementary base pairing **between** the remaining single stranded F-plasmid in **donor/F+ cell** with **free deoxyribonucleotides** to synthesise back the **double stranded F-plasmid** (via the rolling-circle mechanism) ;
- 23. Ref.** to role of CBP for process above – to ensure **accurate transmission** of genetic information in F plasmid;

REJECT - Process:

- *Post transcriptional modification / splicing (Not in prokaryotic cells)*
- *RNA → DNA (Reverse transcription) by reverse transcriptase (Not in prokaryotic cells but in HIV)*
- *RNA → RNA by RNA dependent RNA polymerase (Not in prokaryotic cells but in RNA viruses e.g. Influenza)*
- *Polymerase Chain Reaction (PCR)*

QWC [1]: Answers written in a logical and systematic manner ;

With at least 2 points from category **B** + at least 2 points from category **C** ;

Essay 5

(a) Many different polymers are formed in cells. A polymer is a chain of monomers.

Describe the polymerisation of different types of biomolecules in a plant. [12]

A) **Definition**

1. Polymerisation of all the biomolecules occur through condensation ; water is **lost** / **removed**

B) **[Starch]** *[Reject: Glycogen as it is a storage molecule for animals, instead of plants]*

2. formed from many α -glucose monomers linked together via α -1,4 glycosidic bonds
3. forming amylose and amylopectin
4. A α -1,4 glycosidic bond is formed between the OH group of carbon 1 of an α - glucose and OH group on carbon 4 of the **other** α -glucose (do not double award in cellulose – mark point C8)
5. amylopectin is a branched polymer where the branch points are linked by α -1,6 glycosidic bonds
6. results in a **helical** chain / polymer coiled into a helical shape

C) **[Cellulose]**

7. formed from many β -glucose monomers linked together via β -1,4 glyosidic bonds
8. A β -1,4 glycosidic bond is formed between OH group of carbon 1 of an β -glucose and OH group of carbon 4 of the **other** β -glucose (do not double award in cellulose – mark point B4)
9. **Alternate/successive/neighbouring** glucose monomers are being **inverted/rotated 180° relative to one another**
10. Resulting in a polymer which is **straight** / straight chain
11. catalysed by an enzyme known as cellulose synthase (located on the cell membrane of plant cells)

D) **[Protein]**

12. formed from many amino acids linked together via peptide bonds
13. A peptide bond is formed between the carboxylic acid group / (-COOH) of one amino acid and the **amino group / (-NH₂)** of **another** amino acid
14. Catalyzed by peptidyl transferase (of ribosomes) ; during translation

E) **[Nucleic acid]** / **[DNA and RNA]**

15. formed from many **nucleotides** / nucleoside triphosphates linked together via phosphodiester bonds
16. A phosphodiester bond is between the 3'-OH group of one nucleotide and the 5' phosphate group of an **incoming** / **another** nucleotide
17. Catalyzed by RNA polymerase ; during transcription
OR DNA polymerase ; during DNA replication

QWC: Answers written in a logical and systematic manner ;

with at least 2 points from at least 2 different categories

5(b) Discuss how the complementary shapes between biomolecules enable eukaryotic cells to carry out their diverse processes. [13]

A) Introduction

1. Complementary shape matching may occur between **protein and nucleic acids; or between 2 proteins** [Reject: between 2 nucleic acids as this is by complementary base pairing]

B) [Enzyme-substrate]

2. **Regulating enzyme activity** helps to **conserve energy or resources**
3. Substrates have **complementary 3D conformation** to enzyme's active site ; to form an enzyme-substrate complex
4. Lock and key hypothesis ; Shape of active site is **exactly complementary** to shape of substrate
5. Induced fit hypothesis ; Shape of active site is complementary to shape of substrate, but **slight conformational change to fit more snugly**
[NOTE: Phrase the above statement very carefully! The substrate and enzyme are complementary in shape.]
6. Once product is formed ; no longer complementary to active site ; **released** from active site
7. Named example of enzyme , substrate and product e.g. maltase binds and hydrolyzes maltose to α -glucose
8. Shape of activator / non-competitive inhibitor is complementary to shape of **allosteric site** ; binding **changes conformation** of enzyme and active site ; **increasing/decreasing rate** of enzymatic reaction
9. Shape of competitive inhibitor is complementary to shape of active site ; **competes** with substrate for active site and bind at active site ; **decreasing rate** of enzymatic reaction

C) [Photosynthesis / Respiration]

10. Shape of ATP synthase is complementary to shape of ADP and P_i ; resulting in the synthesis of ATP ; to **drive metabolic reactions of the cell**
11. Shape of NADP reductase is complementary to shape of $NADP^+$ and H^+ ; resulting in the synthesis of NADPH ; to be **used in Calvin cycle**
[Accept: Other enzymes with substrates and functions stated]

D) [Transport across membrane]

12. Carrier protein has a binding site complementary in shape to **molecule (allow example e.g. glucose)**; this allows for molecule to be **transported across** cell membrane;
13. Receptor-mediated endocytosis ; binding site of receptor at cell membrane is complementary to shape of molecule ; for the molecule to be **taken into cell**;

E) [Cell signalling]

14. Ligand binding site of receptors embedded in the cell surface is complementary to the **ligand / signal molecule** ; allowing the cell to **respond to external stimuli**
15. Shape of **second messenger/relay protein** is complementary to shape of **effector protein** ; allowing for the **transmission of signal**
[Accept: Any relay proteins, effector proteins, Ras]

F) [Proteins-nucleic acid interaction]

16. Proteins may have DNA binding {domains / sites} that are complementary in shape to **specific DNA sequence**

[DNA replication]

17. Shape of DNA binding domain of helicase is complementary to shape of DNA sequence at **Ori** ; to **initiate DNA replication** at the correct position / allows the separation of two parental strands
18. Shape of DNA binding domain of primase / DNA polymerase is complementary to shape of 5' phosphate group of free (DNA) nucleotide and 3' OH group of adjacent nucleotide (of existing/growing strand), as well as **parental strand** ; to **synthesize primers** / **Formation** of phosphodiester bond between nucleotides of daughter strand
19. Shape of DNA binding domain of ligase is complementary to shape of 5' phosphate group of one Okazaki fragment and 3' OH group of another Okazaki fragment, as well as **parental strand**; to **seal the nick** between DNA fragments during replication

[Accept: Proofreading ability of DNA polymerase]

[Transcription]

20. Shape of DNA binding domain of general transcription factor / activator / repressor is complementary to shape of DNA sequence at promoter / enhancer / silencer ; resulting in increase / decrease rate of transcription
21. Shape of DNA binding domain of RNA polymerase is complementary to shape of 5' phosphate group of free RNA nucleotide and 3' OH group of adjacent nucleotide (of existing/growing chain), as well as **template strand** ; to allow **formation** of phosphodiester bond between nucleotides of mRNA during transcription

[Translation]

22. A/P/E sites of the large ribosomal subunit binds to tRNA via complementary shape; for **translation**;
23. Shape of amino acid + ATP + tRNA is complementary to shape of active site of aminoacyl-tRNA synthetase ; to allow **activation of amino acid**
24. Shape of translational repressor is complementary to shape of mRNA sequence at 5' UTR / 3' UTR ; resulting in increase / decrease rate of translation
25. Shape of peptidyl transferase is complementary to shape of aminoacyl-tRNA ; for the **formation** of peptide bond between amino acids to form polypeptide during translation

26. Shape of release factor is complementary to shape of mRNA sequence at stop codon ; for the **termination of translation**

G) [Haemoglobin]

27. Each subunit of haemoglobin is complementary to the shape of the haem group ;
28. Reference to **cooperative binding** between 4 polypeptides ; to **facilitate efficient binding to oxygen**

H) [Cell-cell adhesion]

29. Membrane **proteins / glycoproteins / glycolipids** of adjacent **cells** may be **attached together** via complementary shape ; for **cell-cell adhesion**

I) [Immunology]

30. Shape of B cell receptor is complementary to shape of antigen ; to **elicit a humoral immune response / activation** of B cell
[Accept: Epitope of antigen]
31. Shape of T cell receptor is complementary to shape of **antigenic peptide** on MHC of antigen presenting cell / APC (including B cell) ; allowing for **proliferation / activation** of T / B OR formation of memory T / B cells
[Accept: NK cells]
32. Shape of antigen-binding site / variable region of antibodies is complementary to shape of antigen ; for Opsonization / Agglutination / Neutralisation of toxins / Complement activation / Antibody-dependent cytotoxicity
33. Shape of antigen in vaccine is complementary to shape of receptors on **immune cells** ; to **elicit an immune response**

H) [Cell Cycle]

34. **Attachment** of spindle fibre to kinetochore on chromosome ; allows for the arrangement of chromosome in metaphase / separation of chromosome or chromatids in anaphase
35. Shape of CDK is complementary to shape of cyclin ; for the **stimulation of cell division**

36. AVP

QWC [1]: Answers written in a logical and systematic manner ;

with at least 1 correct point from at least 4 categories.